Lab 3 - Loft - Mini Plant Pot with Custom Rails

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 01: Overview of Additive Manufacturing and 3D Scanning TSC objective: A1 - apply the loft technique with guide rails to create a freeform printable vessel

Goal

Model a mini plant pot in Fusion 360 by lofting between a square top and a circular base with custom guide rails, then slice and 3D print it.

What you'll build

A mini plant pot lofted from circle to square, 3D printed without supports.

Tools: Fusion 360, Ultimaker Cura, FDM printer (PLA)

Steps

1. Sketch a 50 mm circle on the XY plane - this is the pot base.

```
Create Sketch > Circle - Ø 50 mm > Finish Sketch
```

2. Create an offset plane 60 mm above the base and sketch a 60 mm square on it.

```
Construct > Offset Plane - 60 mm > Sketch > Rectangle 60 x 60 mm
```

3. Loft from the circle to the square; add guide rails between matching points to control the curve.

```
Create > Loft - profiles: circle + square, rails: optional splines
```

4. Shell the solid to a 2.5 mm wall, removing the top face, to hollow the pot.

```
Modify > Shell - face: top, thickness 2.5 mm
```

5. Export the pot as an STL file.

```
File > Export > STL
```

6. Slice in Cura standing base-down; no supports are needed because the walls flare outward.

```
Cura > Open File > pot.stl > Slice
```

7. Print the pot and inspect the loft surface finish.

```
Cura > Save to Disk > print
```

Evidence to capture

- Fusion 360 screenshot showing both loft profiles, guide rails and the 2.5 mm shell.
- Cura Preview screenshot confirming base-down, support-free orientation.
- Photo or inspection note on wall continuity and the circle-to-square transition.

Test it

The pot prints support-free with an even 2.5 mm wall; the loft transition from circle to square is smooth with no layer separation.